

Examples of Large Green Commercial Buildings in India

- *Covestro's Eco-Commercial Building at Greater Noida*: LEED Platinum rating with 70 per cent reduction in energy usage
- *Suzlon One Earth at Pune*: First building in India to become LEED-certified
- *Infosys Campus at Mysore*: Up to 40 per cent energy and resource savings
- *Confederation of Indian Industry (CII), Sohrabji Godrej Green Business Centre at Hyderabad*: First building outside the US to receive LEED Platinum rating with in-house waste recycling facility
- *Infinity Benchmark building at Kolkata*: Seventh building in the world to receive Platinum LEED rating



Covestro building at Greater Noida

save energy.

"The concept of radiant cooling is becoming popular for comfortable indoor temperature conditions.

"All these solutions are incorporated in buildings to make these sustainable, financially profitable and environment friendly," Dehadani adds.

Many such solutions can also be retrofitted in

existing buildings, which were initially not constructed as green architectures. With some upfront investment, these can be made environment friendly with reduced operational costs.

Investment and payback

While the cost of investment on constructions vary widely according to the scale of the project, it is estimated that a green building at present is at least three to five per cent costlier than a normal building.

Dehadani says, "When green buildings started to develop in India about 10 years ago, upfront capital for green buildings was at least 10 to 12 per cent higher as compared to conventional buildings of equivalent proportions. Now, there is a proper expanding market for these materials, and their prices are becoming reasonable. This has reduced the differential in investments, which can start from a level of five per cent price gap. If green building implementation keeps following this progressive path, the cost will be at par with regular ones."

Various cost savings (on electricity, water, etc) and reduced maintenance hassles ensure payback on excess investments. The payback period may start from the fourth or fifth year. It also depends on the scale of project and investment made.

Evaluating green buildings: certifying bodies

India has several councils and agencies that evaluate green buildings right from their construction phase up to the point where these are being used. Each council has its own process of registration, fee structure, evaluation parameter and rating system, all of which can be explored from their respective websites.

Usual process flow includes online registration, parameter assignment for various evaluations, document submission and on-site evaluation. Ratings can be based on stars or given as platinum, gold, silver, etc.

The major agencies are:

LEED. Leadership in Energy and Environmental Design (LEED) Council's prestigious certification is internationally-recognised and provides a benchmark for high-performance buildings.

IGBC. A division of Confederation of Indian Industry (CII), IGBC works with the government to promote green building construction. It provides national-level ratings to registered projects. IGBC has an evaluation basis for existing building maintenance as well.

GRIHA. GRIHA is a council jointly formed by Ministry of New and Renewable Energy (MNRE) and The Energy and Resources Institute (TERI). It provides in-depth, phase-wise assessments of green building projects that are registered.

GRIHA has also created a free-to-refer catalogue on its website that verifies various products available in the market claiming to be green materials. This is to help consumers identify materials that are authentically environment friendly and suitable for green constructions, and make informed purchases.

As per IGBC reports, India already has more than three billion square feet (68871-acres) of registered green building footprint, which is expected to reach 10 billion by 2022. Costs of green buildings are expected to further drop in this duration, which promises mainstream green construction adaptation in the near future. These are necessary times for adopting such measures, and active cooperation can lead us to a greener tomorrow. **EFY**

GRIHA Council tells us that there is a lot of emphasis on initial construction material management—how much material is being used and in what manner it is being used. There are specific choices of materials. For example, hollow bricks for walls can be used better, as these are lighter and are good insulators of external heat. This keeps the indoors cooler than outdoors.

"Solar panels are quite popular for using renewable energy and cutting down grid energy usage. These greatly save cost of electricity in the long run for the user. Collateral benefits include reduced carbon footprint.

"Water fixtures in green buildings use faucet aerators, which mix a little air in the water stream to create perceived higher pressure. This cuts the water usage down by almost one-third.

"Sensorised water taps in commercial and public buildings can prevent unmonitored open taps.

"Thermal-regulated materials like polycarbonates and hollow bricks help keep indoors cool and comfortable. This improves the living conditions as well as reduces the use of appliances like air-conditioners. This collaterally reduces energy costs and carbon emissions.

"More energy-efficient HVAC systems like variable resistant volume (VLV) or variable resistant flow (VRF) technologies are also being applied to

